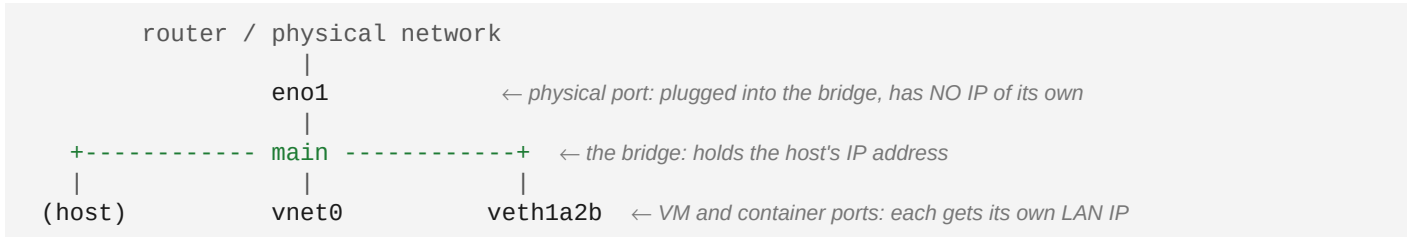


Linux Bridge Cheat Sheet

A bridge is a virtual network switch inside the server. Physical ports and VM/container ports plug into it.

What it looks like



Inspecting a bridge

Command	Shows
<code>ip link show type bridge</code>	Every bridge on the system
<code>ip -d link show main</code>	Confirms main is a bridge (look for a line starting bridge)
<code>ip link show master main</code>	The ports plugged into main
<code>bridge link</code>	Every bridge port, which bridge it belongs to, and its state
<code>bridge fdb show br main</code>	Which MAC addresses the bridge has seen on which port
<code>bridge vlan show</code>	VLANs allowed on each port (if VLAN filtering is on)

Reading bridge link

```
2: eno1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 master main state forwarding
7: vnet0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 master main state forwarding
```

master main = the bridge this port belongs to · **state** = whether the port is passing traffic (see below)

Port state	Meaning
forwarding	Normal: the port is passing traffic
disabled	The port is down (unplugged or switched off)
listening / learning	Starting up (only with STP on); traffic flows within ~30 seconds
blocking	STP has shut the port to prevent a network loop

Reading bridge fdb

```
3c:ec:ef:aa:bb:cc dev eno1 master main ← a device out on the LAN, reached through eno1
52:54:00:12:34:56 dev vnet0 master main ← a VM, reached through vnet0
3c:ec:ef:12:34:56 dev eno1 vlan 1 master main permanent ← the bridge's own address (never expires)
```

This is the bridge's address book. Entries without **permanent** are learned from traffic and expire after about 5 minutes of silence.

Building one by hand (needs sudo; lost on reboot)

Command	Does
<code>ip link add main type bridge</code>	Creates the bridge
<code>ip link set eno1 master main</code>	Plugs eno1 into it
<code>ip link set main up</code>	Switches the bridge on
<code>ip addr add 192.168.1.50/24 dev main</code>	Puts the host's IP on the bridge (remove it from eno1)
<code>ip link set eno1 nomaster</code>	Unplugs eno1 from the bridge
<code>ip link del main</code>	Deletes the bridge

Remember: the IP belongs on the **bridge**, not the physical port. Moving **eno1** into a bridge over SSH cuts you off, so use a console. Make it permanent in Netplan, NetworkManager, or `/etc/network/interfaces`.